[Total No. of Question -04] [Total No. of Pages - 01]

G H Raisoni International Skill Tech University, Pune

T.Y .B.Tech (TERM -I)

CAE I- Winter Term 2026 (2025 Pattern)

Foundations of Data Science (25BTAIMDL507)

(Time: - 01 Hour) (Max. Marks: 20)

Instructions to the students:

i) All questions are compulsory. ii) Neat diagrams must be drawn wherever necessary.
iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed
v) Assume suitable data, if necessary.

Q.No	*Question*	Marks	CO	BL
Q1. (a)	Define Data Science. Explain the need for and importance of Data Science in today's world.	[5]	CO1	2
(b)	Describe the complete Data Science Life Cycle with suitable examples.	[5]	CO1	2
	OR			
Q 2. (a)	Compare and illustrate Business Intelligence (BI) and Data Science. Give at least six differences.	[5]	CO1	4
(b)	Discuss various applications of Data Science in healthcare, finance, education, agriculture, and e-commerce.	[5]	CO1	2
Q 3. (a)	What is Data Transformation? Explain different data transformation techniques.	[5]	CO2	2
(b)	Solve the following: Dataset contains the values: **2, 10, -8, 20, 25, -3, 5** Calculate: • Mean • Median • Mode • Range • Standard Deviation	[5]	CO2	3
	OR			
Q. 4 (a)	Describe various measures of Variability and their significance. Explain each with example.	[5]	CO2	2
(b)	Explain different types of data with suitable examples.	[5]	CO2	2

----------ALL THE BEST----------